BHAVYA PIDIKITI

AI Data Engineer | Cloud Data Engineering

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PROFESSIONAL SUMMARY

Results-driven AI Data Engineer with 5+ years of experience designing scalable cloud data platforms, distributed data processing pipelines, enterprise data warehouses, and production-oriented AI infrastructure across GCP, Azure, and AWS. Proficient in Python, SQL, Apache Spark, PySpark, Databricks, Snowflake, BigQuery, and Apache Airflow, with expertise in building high-throughput ETL/ELT pipelines, medallion architecture data lakes, and Retrieval-Augmented Generation (RAG) systems. Experienced in cloud data migration, SQL query optimization, batch and streaming data integration, automated data quality validation, and CI/CD for data pipelines. Proven ability to optimize processing performance, improve data reliability, and deliver analytics-ready datasets supporting research, retail inventory, supply chain operations, and enterprise decision-making. Skilled in collaborating with engineering, analytics, and machine learning teams to develop scalable, reliable data solutions.

TECHNICAL SKILLS

AI Data Infrastructure & LLMOps: Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, LangChain, LlamaIndex, Vector Databases, Feature Stores, LLM Ingestion Pipelines, RAG Data Pipelines, Retrieval Optimization, AI Data Preparation

Data Engineering & Orchestration: Apache Airflow, Apache Kafka, Apache Spark, PySpark, Delta Lake, dbt, ETL/ELT Pipelines, REST APIs, Dimensional Modeling, Star Schema, Batch & Streaming Data Processing, Cloud Data Migration, Data Pipeline Architecture, Cross-Source Data Integration

Cloud Platforms & Data Services: Google Cloud Platform (GCS, BigQuery, Cloud Composer), Microsoft Azure (Azure Synapse Analytics, Azure DevOps, Azure Functions), AWS (S3, Glue, Redshift, EMR), Snowflake, Databricks, PostgreSQL, NoSQL, Cloud Data Warehousing, Lakehouse Architecture, Enterprise Data Platforms

Programming & Big Data: Python, Advanced SQL, T-SQL, JSON, Parquet, Docker, Distributed Computing, SQL Query Optimization, Data Preprocessing, Large-Scale Data Processing

Data Quality & CI/CD: GitLab CI/CD, GitHub Actions, Jenkins, Azure DevOps, Schema Validation, Data Governance, Automated Testing & Quality Checks, Data Quality Automation, Data Contracts, Data Validation Frameworks, CI/CD for Data Pipelines

PROFESSIONAL EXPERIENCE

Penn State University **USA | May 2025 – Present**

AI Data Engineer

- Optimized PySpark data pipelines processing student and research text logs, automating Delta Lake ETL stages and feeding vector embeddings into RAG frameworks to reduce weekly processing time by 15%.
- Engineered SQL and distributed ETL pipelines indexing campus feedback into a localized retrieval-augmented generation framework, resolving partitioning bottlenecks to improve retrieval precision by 12% and support search portals handling 5K queries.
- Built automated Python and Apache Spark workflows extracting interaction logs into a medallion architecture data lake, standardizing data transformations and generating clean text embeddings for semantic search and academic advisory applications.
- Managed migrating legacy relational databases into a centralized PostgreSQL warehouse via Apache Airflow workflows supporting a 10-person research analytics team, unifying student engagement datasets for downstream analytics and LLM retrieval pipelines.
- Implemented multi-layered automated validation checks across staging layers for campus surveys and audio transcripts, detecting malformed payloads before downstream processing, saving 8 manual hours weekly, and strengthening research data quality.
- Collaborated with machine learning researchers to design data contracts and GitLab CI/CD routines for natural language pipelines, standardizing ingestion processes and supporting reliable deployment of research-oriented AI data workflows.

The Kroger Co. **USA | Aug 2024 – Apr 2025**

Data/AI Engineer

- Built Apache Spark ETL workflows in Databricks to process live inventory data for 84.51° analytics models, creating standardized feature tables and eliminating stale inventory records to improve downstream merchandising analytics and AI data reliability.
- Developed event-driven Kafka streams capturing Point of Sale transactions for the Kroger Plus customer loyalty program, optimizing topic partitioning to reduce downstream processing latency by 18% and support real-time personalization models.
- Transitioned on-premises Hadoop data into Google Cloud Storage and BigQuery using Delta Lake, scheduling batch workloads via Cloud Composer to modernize retail data infrastructure and support scalable cloud-based analytics and AI processing.
- Engineered a modular e-commerce data architecture using dbt and standardized SQL macros, developing reusable dimensional models and transformation logic to improve data consistency, pipeline maintainability, and retail analytics accessibility.

- Integrated schema validation checks into GitLab CI/CD pipelines to verify daily pricing updates, preventing invalid promotional records from reaching downstream data systems and reducing monthly data support tickets by 32%.
- Deployed curated data views connecting Snowflake warehouses directly to BI dashboards and feature stores, transforming nested JSON datasets into analytics-ready relational structures for merchandising analysis and predictive modeling.

Microsoft **India | Jun 2021 – Aug 2023**

Data Engineer

- Engineered end-to-end data ingestion pipelines integrating legacy relational databases and high-throughput NoSQL stores into centralized cloud data lakes, standardizing enterprise data structures and improving consistency across global financial transaction datasets.
- Optimized large-scale data processing workloads using Azure Synapse Analytics and distributed Spark clusters, accelerating SQL query execution by 3x and improving analytical performance across high-concurrency enterprise environments.
- Built comprehensive data preprocessing and automated validation workflows using Python to clean incoming telemetry feeds, handling missing values, standardizing data formats, and improving overall dataset reliability by 25%.
- Designed enterprise data warehouse structures using dimensional modeling and optimized star schemas to unify operational data sources, reducing data retrieval latency by 18% and improving accessibility for downstream business intelligence and analytics.
- Developed interactive, publication-grade Power BI dashboards and reporting frameworks, enabling 20+ stakeholders to track operational KPIs, analyze trends, and make informed business decisions using centralized reporting datasets.
- Automated background data processing pipelines using serverless computing frameworks and established CI/CD workflows with Azure DevOps, standardizing cloud deployment processes and reducing manual configuration dependencies across environments.

Data Analyst Intern **India | Jun 2020 – May 2021**

- Wrote optimized baseline SQL queries and cleansed raw data exports within Azure Databricks notebooks to support exploratory data analysis for comprehensive reporting deliverables across 3 active accounts, ensuring high-quality datasets.
- Supported the validation testing phase of enterprise ETL migration tasks by running initial row-count checks, identifying 12 recurring data mapping discrepancies across staging tables and improving data accuracy by 14%.
- Collaborated with analytics teams to construct 6 interactive Power BI dashboards and update internal technical documentation, building foundational experience in data warehousing, SQL-based analysis, data visualization, and dimensional modeling.

KEY PROJECTS

Real-Time Fraud Detection & Transaction Risk Scoring Pipeline

Technologies: *Python, Apache Kafka, PySpark, AWS Lambda, Amazon Redshift, Amazon SageMaker*

- Integrated a SageMaker-hosted anomaly detection model into the streaming pipeline, preparing transaction-level features including transaction velocity, geolocation mismatch, and spending pattern deviation for real-time risk scoring and fraud analysis.
- Built serverless AWS Lambda functions to trigger automated risk alerts and block flagged transactions when anomaly scores crossed critical thresholds, supporting real-time transaction monitoring and an estimated 25% reduction in fraudulent exposure.

Multi-Vendor Retail Supply Chain & Inventory Optimization Platform

Technologies: *Python, Snowflake, dbt, Apache Airflow, Tableau*

- Designed and implemented an automated inventory synchronization platform aggregating multi-warehouse stock levels, vendor fulfillment logs, and real-time shipping data into a centralized Snowflake cloud data warehouse, improving cross-source data integration and inventory visibility.
- Built modular dbt transformation models and automated Airflow data pipelines to calculate dynamic safety stock thresholds, enabling inventory optimization and contributing to an 18% reduction in excess inventory holding costs.

EDUCATION

Master of Science in Information Systems Aug 2023 – May 2025
University of Cincinnati | Cincinnati, Ohio

Bachelor's in Information Technology Jul 2017 – Jul 2021
GITAM University | Visakhapatnam, Andhra Pradesh

CERTIFICATIONS & TRAINING

Microsoft Certified: Azure Data Engineer Associate
AWS Certified Cloud Practitioner
Google Data Analytics Professional Certificate